

# Build & Flash Guide

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This guide covers downloading the Flight Radar code and uploading it to your Waveshare ESP32-S3-Touch-LCD-1.46C board.

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## Quick Start (For the Impatient)

1. **Download:** <https://github.com/computethis/TheMakersSpace-Flight-Radar> → Code → Download ZIP
2. **Extract** to Desktop → Rename folder to **FlightRadar**
3. **Open:** **FlightRadar/src/FlightRadar/FlightRadar.ino** in Arduino IDE
4. **Install Libraries:** Arduino\_GFX\_Library (1.5.7+), ArduinoJson (7.x), WiFiManager (2.0.x), SensorLib
5. **Board Settings:** Select "Waveshare ESP32-S3-Touch-LCD-1.46", PSRAM: **Enabled**
6. **Upload** → Verify display shows "SETUP"

**Stuck?** See the detailed steps below.

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## Part 1: Download the Code

### Option A: Download ZIP (Recommended)

1. Go to <https://github.com/computethis/TheMakersSpace-Flight-Radar>
2. Click the green "<> Code" button
3. Select **"Download ZIP"**
4. Extract the ZIP file to your Desktop
5. **Rename** the extracted folder to **FlightRadar** (removes the **-main** suffix)

You should now have:

```
Desktop/
├── FlightRadar/
│   ├── src/
│   │   └── FlightRadar/
│   │       ├── FlightRadar.ino    ← Open this
│   │       ├── board_config.h
│   │       ├── aircraft_types.h
│   │       ├── themes.h
│   │       ├── ui_layout.h
│   │       └── font_config.h
│   ├── docs/
│   └── README.md
```

### Option B: Use Git (If Installed)

```
cd Desktop
git clone https://github.com/computethis/TheMakersSpace-Flight-Radar.git
```

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## Part 2: Install Arduino IDE & ESP32 Support

### First-Time Setup

If you already have Arduino IDE with ESP32 support, skip to Part 3.

1. Download Arduino IDE from <https://www.arduino.cc/en/software>
  - Version 1.8.x or 2.x both work
2. Install and launch Arduino IDE
3. Open **File** → **Preferences** (Arduino → Preferences on Mac)
4. Find "Additional Board Manager URLs" and add:
  1. [https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package\\_esp32\\_index.json](https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json)
    - If there's already a URL there, click the icon and add this on a new line
5. Click **OK**
6. Go to **Tools** → **Board** → **Board Manager**
7. Search for "**ESP32**"
8. Install "**ESP32 by Espressif Systems**" (latest version)
9. Wait for installation to complete (may take a few minutes)

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## Part 3: Install Required Libraries

Go to **Sketch** → **Include Library** → **Manage Libraries...**

In the Library Manager search box, install each of these:

| Library                             | Version         | Author          |
|-------------------------------------|-----------------|-----------------|
| <a href="#">Arduino_GFX_Library</a> | 1.5.7 or higher | moononournation |
| <a href="#">ArduinoJson</a>         | 7.x             | Benoit Blanchon |
| <a href="#">WiFiManager</a>         | 2.0.x           | tzapu           |
| <a href="#">SensorLib</a>           | Latest          | Lewis He        |

### Installation tips:

- Search for the exact name shown above
- Click **Install** for each library
- If you see multiple results, pick the one with the matching author

- Some libraries may already be installed— that's fine!

**Close and restart Arduino IDE** after installing libraries.

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## Part 4: Open the Project

1. In Arduino IDE: **File** → **Open**
2. Navigate to `Desktop/FlightRadar/src/FlightRadar/`
3. Select `FlightRadar.ino`
4. Click **Open**

You should see:

- Main code in the editor
  - Tabs at the top: `FlightRadar.ino`, `board_config.h`, `aircraft_types.h`, `themes.h`, `font_config.h`
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## Part 5: Configure Board Settings

Connect your board to the computer with a USB-C **data cable** (not a charge-only cable).

### Select the Board

Go to **Tools** menu and set **exactly** these options:

```
Board:           Wavesheare ESP32-S3-Touch-LCD-1.46
USB CDC On Boot: Enabled
Events Run On:   Core 0
Flash Mode:      QIO 120MHz
Arduino Code Runs On: Core 1
Partition Scheme: 8M with spiffs (3MB APP/1.5MB SPIFFS)
PSRAM:           Enabled ← CRITICAL!
Upload Speed:    921600
USB Mode:        Hardware CDC and JTAG
```

 **Critical:** PSRAM must be set to **"Enabled"** (not "Disabled"). The display framebuffer requires PSRAM. If PSRAM is disabled, the screen will stay black.

### Select the Port

Go to **Tools** → **Port** and select:

- **Windows:** `COM3` or similar (may show "USB-SERIAL")
- **Mac:** `/dev/cu.usbserial-XXXX` or `/dev/cu.wchusbserial-XXXX`
- **Linux:** `/dev/ttyUSB0` or similar

### If no port appears:

- Try a different USB cable (must be data cable, not charge-only)
  - Install CP210x USB driver (Windows): <https://www.silabs.com/developers/usb-to-uart-bridge-vcp-drivers>
  - On Mac, wait 30 seconds after plugging in—ports take time to appear
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## Part 6: Upload the Code

Click the → **(Upload)** button or select **Sketch** → **Upload**

### If Upload Fails

#### Error: "Failed to connect to ESP32"

Use the BOOT button method:

1. Hold down the **BOOT** button on the board (small button labeled "BOOT")
2. While holding, click **Upload** in Arduino IDE
3. Keep holding until you see "Connecting..." in the output window
4. Release the BOOT button
5. If it still fails, try again

#### Error: "A fatal error occurred: Failed to write to target RAM"

- Press the **RST** button on the board
- Try upload again with BOOT button held

**Success:** You should see:

```
Leaving...  
Hard resetting via RTS pin...
```

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## Part 7: Verify It Works

### Check the Serial Monitor

1. Open **Tools** → **Serial Monitor** (or press Ctrl+Shift+M / Cmd+Shift+M)
2. Set baud rate to **115200** (bottom right dropdown)
3. Press the **RST** button on your board

You should see:

```
[Boot] FlightRadar starting...
[Boot] Waveshare ESP32-S3-Touch-LCD-1.46C
[Touch] SPD2010 reset complete
[IMU] QMI8658 ready
[FS] LittleFS ready – XXXX KB free
[LCD] panel + canvas ready
[WiFi] connected – IP=192.168.X.XXX
[mDNS] flightradar.local
[OTA] ready
[HTTP] web server up on port 80
[Boot] ready
```

## Check the Display

The round LCD should show:

**"SETUP"** in the center with a WiFi name like:

```
FlightRadar-A1B2
```

If you see this, **success!** Proceed to WiFi configuration in the [Student Guide](#).

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## Troubleshooting

| Problem                         | Solution                                                                 |
|---------------------------------|--------------------------------------------------------------------------|
| No port appears                 | Try different USB cable; install CP210x driver (Windows); wait 30s (Mac) |
| "Failed to connect"             | Hold BOOT button during upload                                           |
| "Canvas->begin() FAILED"        | Check PSRAM is set to <b>Enabled</b> in Tools menu                       |
| Upload hangs at "Connecting..." | Press RST, then retry with BOOT button held                              |
| Display stays black             | Press RST button; verify PSRAM setting                                   |
| "Library not found"             | Check spelling; install via Library Manager                              |
| Serial Monitor shows garbage    | Set baud rate to 115200, not 9600                                        |
| Touch doesn't work              | Restart board; SPD2010 needs a few seconds after boot                    |

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## Next Steps

- **Students:** Return to [Student Guide](#) for WiFi configuration

- **Instructors:** See [Leader Guide](#) for teaching notes and troubleshooting tips
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## Advanced: Flashing Without USB (OTA)

Once the board is running and connected to WiFi, you can flash updates wirelessly:

1. In Arduino IDE: **Tools** → **Port**
2. Look for **flightradar** at **192.168.X.XXX** (network port)
3. Select it
4. Click Upload as normal

No USB cable needed! Requires Arduino IDE and board on same WiFi network.